

1

See zip

Vector1 has a length of 100,000

2

See zip

Vector2 has a length of 100,000

3

See zip

Vec1 first 10: -7.36924, -0.826997, -5.62082, 3.57729, 8.69386, 0.388327, -9.30856,
0.594004, -9.84604, -8.6 6316

Vec2 first 10: -3.27266, -9.31481, 6.54747, -8.61734, 0.650344, 5.80275, 9.91973,
2.77242, 8.26379, -3.78691

4

The difference between vectors and arrays in C++ is that vectors are resizable while arrays are not.

5

See zip

Written tto data/hadamard.txt

6

See zip

Same with a tolerance of 0.0001 which I attributed to rounding errors

7

I had very basic exposure to C++ like 7 or 8 years ago and have not used it since, so I have very little C++ experience. The exercises helped me remember coding in languages like C (which has also been around 3 years now) as I have almost exclusively coded in python for the past few years.